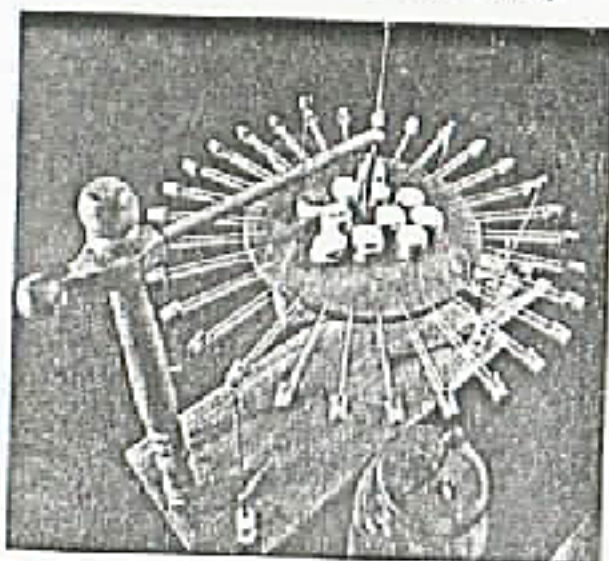


Fig. 1. A replica of Franklin's electrostatic motor. Rotation is achieved by corona discharge.



ELECTRIC POWER FROM THE EARTH

THE SEARCH FOR NATURAL SOURCES OF ENERGY GOES ON

BY L. GEORGE LAWRENCE

MOST economists agree that present standards of living can be maintained only if more electrical power is made available every year. Unfortunately, to obtain power from traditional sources means pollution; getting more power from these sources means more pollution. Consequently, it comes as no surprise that a great deal of effort and capital are being invested in attempts to find "cleaner" sources of electricity than we have been using. Along these lines, we find renewed interest not only in geoelectric phenomena, but also in exotic energy conversion schemes with excellent promise for the not-too-distant future. So far, everything has been very experimental, but, at last, a beginning has been made.

To geophysicists and electrical engineers, our planet is a powerful generator of dc power in the trillion-kilowatt (10^{12}) range. Its dynamo action is due partially to axial rotation and the magnetic interaction in ionospheric fields and partially to thermocouple-type phenomena between the hot

magma of the earth's core and the cold crust. The natural earth, or telluric, currents flow in large "sheets." While apparently weak, telluric currents can attain immense magnitudes in submarine telephone cables when certain geoelectric conditions coincide. (Submarine cables between the U.S. and Europe frequently produced terminal voltages up to 2500 volts.)

The question is, how can we harness this power? Studies are currently under way to determine how to increase the current flow in ore-bearing bodies and store the electrical energy potentials in buffers or accumulators for later use. Special provisions must be made for possible telluric polarity changes to safeguard dc-to-ac converter systems.

Another force that is available for immediate use is that of electrostatic energy. Electric fields, currents, and conductivity, as well as positive and negative ions of greatly varying size, constitute the principal electrical properties of the atmosphere in fair weather. In the altitude between 30 and

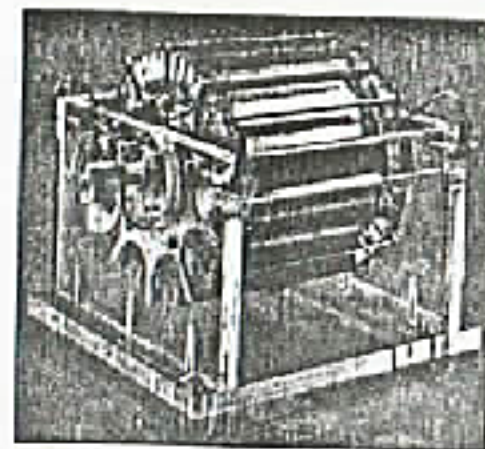


Fig. 2. The electrostatic motor shown here can develop 1/10 horsepower by using an antenna carried by a balloon.

70 kilometers, atmospheric studies of electricity and ionospheric physics merge.

Remarkably enough, Benjamin Franklin was among the very first people to envision the enormous potential of static electricity in industry. An exact replica of his electrostatic corona motor, recreated at West Virginia University by Dr. Oleg Jefimenko and David K. Walker, is shown in Fig. 1. The propulsive force for the motor disk is obtained from discharge electrodes placed at strategic points around the disk's circumference. A Leyden jar (lower right in

photo) serves as the high-voltage storage/discharge component.

The latest version of the WVU electrostatic motor is shown in Fig. 2. Like similar machines, the insulated knife-edge electrodes generate a corona that charges or ionizes air particles floating by. The latter transfer their charge to the nearest part of the plastic rotor and charge it. Now, once a spot on the rotor takes a charge, it will be repelled by simple electrostatic force from the charging electrode. A simultaneous attraction toward the other, oppositely-charged, electrode takes place. So, when the charged portion of the rotor reaches the opposite electrode, another corona discharge takes place, reversing the polarity, and the cycle repeats itself.

The Jefimenko model shown can develop 0.10 horsepower if energized by a 30,000-volt electrostatic generator. Operation by natural electrostatic electricity requires that the feeder antenna be carried aloft by kite or balloon to a height of 200 to 300 ft above ground to obtain an operating potential of about 20,000 volts. Of course, all of this sounds promising, but our society is in a hurry and needs large amounts of power now.

More Practical Approaches. Presently, considerable attention and a great deal of money are being used to exploit hot springs and natural geysers as sources of power.

Fig. 3. Direct-conversion geothermal power system.

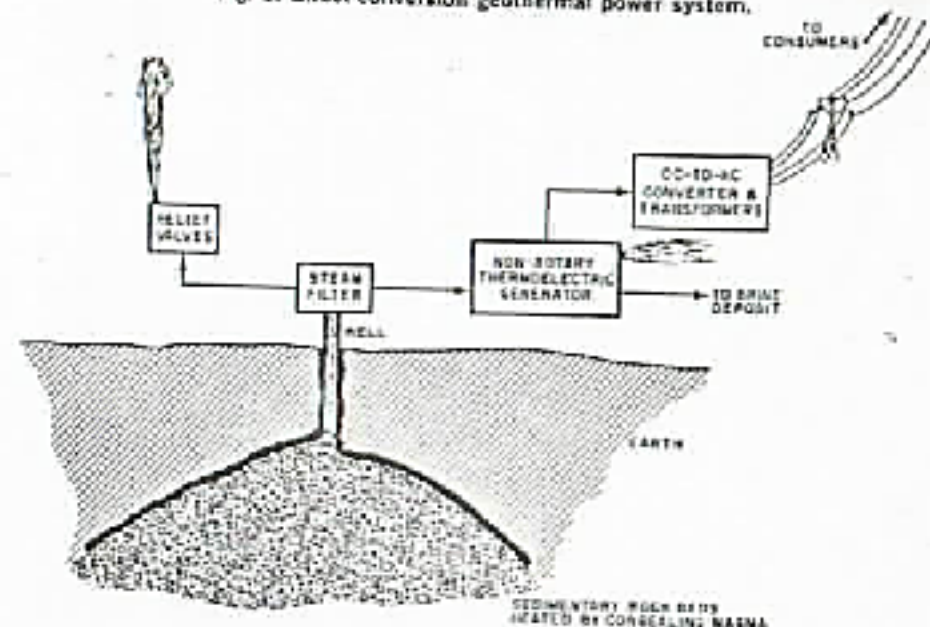




Fig. 4. Model car made by Huffman is powered by a thermotunnel converter.

One promising spring is at Cerro Prieto in northern Mexico, where Toshiba of Japan has installed specially designed turbines to convert escaping steam into electrical power. But the turbine generator conversion process is not ideal; more direct heat-to-electricity conversion processes are being developed.

Typically, a geothermal spring is made up of a vast underground deposit of spongy sedimentary rock (see Fig. 3) that traps water that has been heated to the boiling point by heat conduction from the solid rock below. The heat is furnished by congealing magma pushed up from the earth's molten core. Geologists have estimated that, in the case of the Mexican and Salton-Sea trough systems, spongy rocks to a depth of 20,000 ft exist, giving an immense energy reservoir. However, in the system concept shown, it is desirable that turbines and the like are avoided altogether either by tapping the heat directly (to avoid turbine contaminations) or by feeding the hot steam into a non-rotary thermoelectric generator as shown. Once fully developed, direct heat-to-electricity converters are cheaper than turbine generators and offer fewer logistical problems in the long run.

Right now, although good progress is being made, problems reside in the area of efficient conversion cells. This field is wide open for new inventions in light of the fact that the need is one of here and now.

A small experimental car, about 8-in. long and containing a small electric motor, is driven by an electronics-based thermotunnel converter invented and patented by F.N. Huffman (Fig. 4). The converter uses 10-40-angstrom-thick oxide spacers between the emitter and collector of the heat-to-electricity conversion assembly. Buffer resistors are inserted in the device to insure an even load balance from all current-generating cells during parallel operation.

While at first glance, Huffman's invention (little more than a sophisticated thermocouple) seems insignificant, it bears great promise for large-scale power plants. What remains is essentially an engineering job of finding a packaging scheme that will permit this type of cell to be directly heated by the earth's magma or steam from hot springs.

Work is also under way in the field of atomic power plants. The specific aim is to bypass the traditional nuclear power plant and its cumbersome reactors, turbines, and the like by realizing the current-generating capacity of the radioactive materials themselves. In Fig. 5 are shown two "classical" nuclear battery types. The design in (A) is a beta-current type in which high-speed electrons emerge from the radioisotope coated on a central emitter electrode and are collected by an outer conducting member. Design (B), coated on one side with Strontium-90, allows the radiation to produce an electron avalanche in one element of the junction diode. These electrons are collected

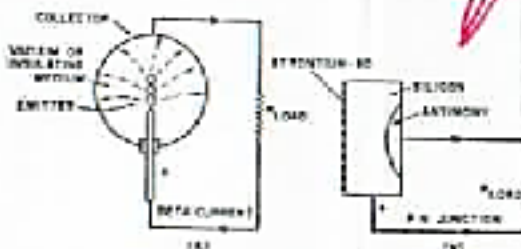


Fig. 5. Schematics of two classical battery types using nuclear energy.

by the "field" across the junction. The field is established by a physical process similar to the collecting field in the contact-potential-difference cell.

Unfortunately, these two and similar types of batteries produce very low output powers, typically in the microwatt range. However, cell (B) continues to be promising because its electromotive force is about 0.2 volt per cell. So, if a large number of radiation-proofed cells can be connected in series-parallel combinations, adequate power outputs might be realized. Current cell failures are due to changes in the crystalline structure of the silicon that are the result of radiation from the radioactive material. But it is this type of solid-state battery that holds the great promise for electric-powered cars, homes, and industrial undertakings. ☐

USING YOUR CB SET FOR EMERGENCIES

Operating rules and regulations
to make channel 9
a more effective emergency
communications network

EVERYONE has his own personal reason for getting into CB radio. You may want to use it for business, or just personal communications within your own family. You may want to talk to friends or meet new friends through your radio. Whatever your original purpose, sooner or later, you will find that your personal two-way radio has a great value for emergency communication—your own or someone else's.

Emergency use of your radio will probably first come about when you sight an accident, fire, or traffic hazard while driving. Of course, all you have to do is call on channel 9, the official emergency channel, to report it. Who will respond? Probably a member of a volunteer emergency radio team or a public spirited individual; and, in rare instances, a police department.

According to the Citizens Radio Section, Electronic Industries Association, over 2,000 volunteer groups conduct organized

monitoring of emergency channel 9. In addition, they estimate that 500 police and fire departments are now monitoring CB.

It is important to know how to call for assistance or to report an emergency on channel 9. If you know the call letters of the local channel 9 monitors, it is simple a matter of calling them as you would be any other call, on any other channel.

When you don't have the call sign of an emergency monitor, as is the case when you are traveling, it is another matter. Try, "KZZ-4321 mobile, seeking assistance." Or, "KZZ-4321 mobile with a 10-33 (emergency)." If you are traveling out-of-state, give your home state as, "KZZ-4321, California mobile." Allow time for response before repeating call. However, don't expect an immediate answer the first time. You may have to wait until the monitor hears you repeat before he can identify your signal. If reporting an accident or other emer-

The EIA highlights emergency uses of CB by means of this public service billboard. It will appear around the country in 1973. Interested CB clubs can get this message posted by having local outdoor advertising firm request material from Citizens Radio Information, 7th Floor, 21 East Hubbard Street, Chicago, Illinois 60611.

No woman should drive alone





**without the safety of
Citizens two-way radio**

— 9 —